

LSTMs, GRUs, and Gating Mechanisms

Scaler School of Technology | Deep Learning I | Week 6 — Session 1/2 | May 2026



What Is This Lecture About?

Last session we saw vanilla RNNs **break** on long sequences due to vanishing gradients. This session introduces the architectures that **fix** it — LSTMs and GRUs use learnable gates to control information flow, keeping gradients alive across hundreds of timesteps. You'll also learn when to use bidirectional and deep (stacked) variants.

1 LSTM Cell Anatomy & Gradient Flow

Three gates (forget, input, output) control a “conveyor belt” cell state that lets gradients flow without decay. *Think of it as a highway with on-ramps and off-ramps — information cruises straight through unless a gate deliberately intervenes.*

2 GRU — The Simplified Alternative

Only 2 gates (reset, update) and no separate cell state. Achieves similar performance with 25% fewer parameters. *Like simplifying a 3-knob shower to a 2-knob one — same hot water, fewer controls.*

3 Bidirectional & Deep RNNs

Bidirectional processes sequences in both directions for full context. Stacking multiple layers captures hierarchical features. Together: the “deep bidirectional LSTM” that dominated NLP from 2015–2017.

What to Expect in the Session

- Build an LSTM cell from scratch in numpy
- Visualize gate activations on patterned sequences
- ★ Race: LSTM vs GRU vs vanilla RNN on a long-range memory task
- Compare parameter counts across architectures
- Implement bidirectional and stacked LSTMs in PyTorch
- Quizzes after each major section

How to Prepare (Optional but Helpful)

- ✓ **Recall** the vanishing gradient problem from last session
- ✓ **Think about:** how might you let gradients “skip” through time? (Hint: ResNet did this for layers)
- ✓ **Review** sigmoid function — outputs 0 to 1, used as a “valve”
- ✓ **Optional:** skim Colah's “Understanding LSTMs” blog post for the diagrams

Duration: ~2 hours

Prerequisites: Vanilla RNN, vanishing gradients, BPTT

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